

A Scoping Review of Epigenetic Signatures of Diet and Diet-related Metabolites: Insights from Epigenome-Wide Association Studies and Their Implications for Cardiometabolic Health and Diseases

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Epigenome-wide association studies (EWASs) have emerged as a powerful approach to investigate how dietary exposures shape the epigenome and subsequently influence metabolic and cardiovascular health. A growing number of EWAS have examined the effects of various dietary factors, including overall dietary patterns, specific food groups, micronutrients, and food-related metabolites, on DNA methylation (DNAm) across diverse populations. In this review, we map the landscape of nutritional EWAS, identifying the types of dietary exposures studied, the genomic regions where epigenetic signals emerge, and overarching trends across studies. Across studies, consistent associations were reported at nine CpG sites in genes such as *AHRR*, *CPT1A*, and *FADS2*, particularly in relation to fatty acid consumption, and certain diet patterns. Biological pathways enriched included fatty acid metabolism and the PPAR signaling pathway. In conclusion, our review identified a pattern of epigenetic convergence that may underlie diet-related disease risk. While promising, key knowledge gaps were also noted, including limited longitudinal follow-up, unclear causal pathways, and underrepresentation of ethnic diversity. Moving forward, we highlighted several complementary approaches for translating nutritional EWAS findings into actionable public health and precision nutrition strategies, including integrating multi-omics, mediation analyses, and population-wide epigenetic risk profiling.

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INTRODUCTION

Diet is a key determinant of health, influencing metabolism, immune function, cognitive performance, and disease risk. Once ingested, certain dietary components are absorbed and metabolized into circulating bioactive compounds [1], which can be detected as metabolic signatures of dietary intake [2]. Growing research supports the hypothesis that foods, including macronutrients, micronutrients, and bioactive food compounds, contribute to the development of metabolic disorders [3,4], cardiovascular disease [5], and other non-communicable chronic diseases [6]. Diet can affect long-term health through multiple mechanisms, such as cumulative dose-dependent exposure and epigenetic modifications [7], which “reprogram” gene expression to regulate metabolism [2], immune responses, disease susceptibility, and healthy aging [7].

Epigenome-wide association studies (EWASs) [8] have emerged as a powerful approach to understanding how environmental exposures, particularly diet, influence epigenetic changes. DNA methylation (DNAm) is one of the most studied epigenetic variations as it can be profiled at scale in large population studies. This scoping review synthesizes EWASs of dietary measures, with a focus on dietary patterns, micronutrients, macronutrients, and diet-related metabolites. The objectives are to: 1) characterize the dietary outcomes studied in EWASs; 2) identify overlapping epigenetic signals from dietary and related metabolite EWASs; and 3) evaluate whether epigenetic signals of dietary exposures influence long-term health outcomes.

Dietary Patterns and Measures

Diet is a complex and dynamic phenotype influenced by multiple biological, behavioral, and environmental factors, making it challenging to accurately capture in research. It can be classified qualitatively into patterns—eg, Mediterranean, or plant-based diets—and quantitatively by estimating intake of nutrients and food processing levels [9,10]. Accurate measurement of diet enables precision nutrition through tailored dietary recommendations based on genetics, metabolism, epigenetics, and microbiota [11]. Additionally, accurate measurements are crucial for understanding lifelong dietary exposures that shape metabolism and disease risk. Traditional dietary assessment methods have well-described limitations, such as misreporting, social desirability bias, and recall bias. While digital tools offer potential improvement, their validity across diverse populations remains a concern.

Metabolomic Signature of Diet

Advances in high-resolution mass spectrometry and

nuclear magnetic resonance spectroscopy [12] have significantly improved metabolite profiling, a more detailed characterization of the metabolome—the complete set of small molecules present within a biological sample [13]. This includes the entirety of low-molecular-weight compounds (<1000 Da) such as amino acids, peptides, lipids, nucleotides, and secondary metabolites that are end products of cellular processes. Metabolites can be measured in serum, plasma, and urine. Blood metabolites reflect hemodynamic regulation, while urinary metabolites reflect renal filtration and excretion; thus, together, they offer complementary health indicators [14].

Diet modulates the metabolome. For instance, a diet high in processed foods influences the fecal metabolome, altering concentrations of specific metabolites such as dipeptides and long-chain saturated fatty acids. These changes reflect altered protein digestibility and nutrient bioavailability resulting from food processing methods [15]. The metabolome also reflects the interactions between ingested foods, their processing, the gut microbiota, and host biology. Early-life dietary patterns, such as breastfeeding versus formula feeding, impact the infant serum metabolome [16,17]. Notable dietary biomarkers include nutrients and metabolite associations, such as dietary fructose, glucose, vitamin C, and proline betaine, a known metabolomic biomarker of citrus fruit intake [18]. Despite clear diet and metabolome connections, it is important to note that metabolomic profiles only partially capture dietary intake. Current studies suggest that the correlation between self-reported diets and metabolomic data tends to be modest, with a correlation coefficient $r = 0.3$ to 0.4 at best [19], depending on dietary assessment methods and individual metabolic variability.

DNA Methylation

Epigenetic regulation includes mechanisms such as DNAm, histone modifications, and non-coding RNAs, all of which influence gene expression without altering the DNA sequence. DNAm is currently the most extensively studied in large population studies. It typically refers to the covalent addition of a methyl group ($-\text{CH}_3$) to the 5-carbon position of the cytosine ring, forming 5-methylcytosine (5mC), predominantly within cytosine-phosphate-guanine (CpG) dinucleotides. This modification can influence gene expression by altering transcription factor binding or recruiting methyl-binding proteins that remodel chromatin. Newer, long-read sequencing technologies now allow for the detection of methylation beyond the 5mC context. However, this review focuses on 5mC, given its established relevance and widespread use in current research. In contrast to genetic alterations, DNAm changes are reversible and sensitive to environmental factors and lifestyle choices [15,16]. Indeed, the

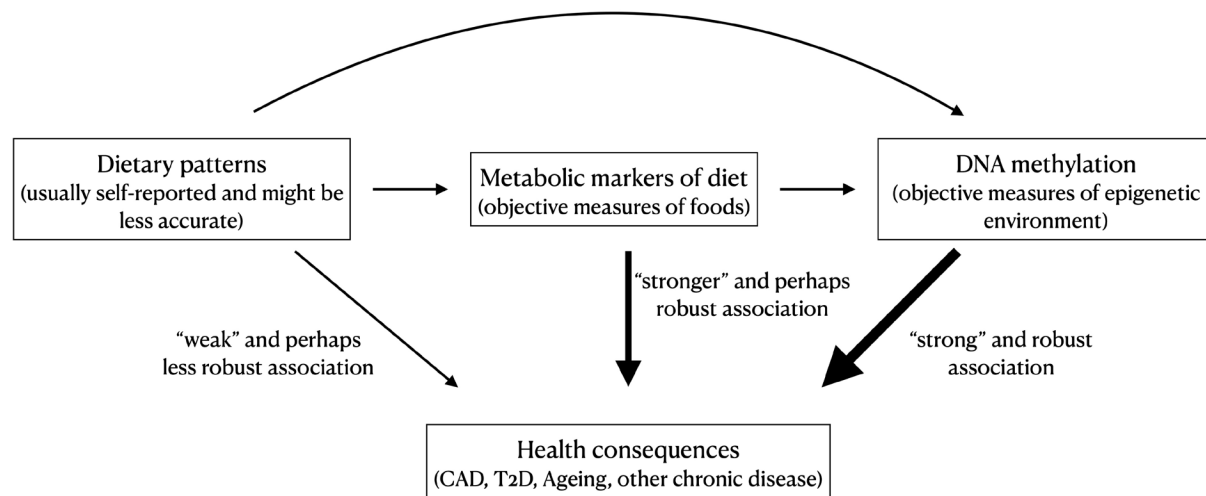


Figure 1. Conceptual framework linking dietary patterns, dietary metabolites, and DNA methylation to chronic disease outcomes. This diagram illustrates the hypothesized relationships between subjective measures of diet (dietary patterns) and more objective biomarkers of intake and exposure (dietary metabolites and DNA methylation) in relation to health consequences such as coronary artery disease (CAD), type 2 diabetes (T2D), aging, and other chronic conditions. We hypothesize that dietary patterns may show weaker and less robust associations with outcomes due to recall bias and imprecision. In contrast, dietary metabolites, as more objective indicators of recent intake, may be more strongly and immediately associated with health outcomes. Finally, DNA methylation captures the long-term biological responses to exposure, potentially reflecting the cumulative effects of diet and other environmental factors. Arrows indicate hypothesized directions of influence, with DNA methylation and dietary metabolites acting as intermediates or biomarkers linking diet to disease.

growing number of EWASs in relation to environmental exposures and health outcomes represents the early stages of a systematic effort to identify epigenetic signatures linked to disease and biological processes.

Modern array-based technologies, such as the Illumina Infinium HumanMethylation450 BeadChip (450K) and its successor, the MethylationEPIC BeadChip (Illumina Inc, San Diego, CA), are commonly used. These platforms interrogate between 450 000 to over 850 000 CpG sites, providing extensive coverage of the methylation landscape. The generated beta-values, which range from 0 (completely unmethylated) to 1 (fully methylated), quantify the fraction of methylation at each CpG site.

Why Diet-Metabolome-Epigenome

Diet can induce reversible epigenetic changes [17,18], prompting interest in capturing dietary signatures relevant to chronic disease prevention and management. A conceptual overview of these interactions is illustrated in Figure 1, highlighting proposed pathways by which dietary exposures influence metabolite profiles and DNAm. Given the aforementioned complexities in nutrition research [19], reliable dietary biomarkers present a promising complement. A dietary biomarker is a measurable biological indicator that reflects dietary

intake, metabolism, or nutritional status [20], and could offer more objective and precise measurements of dietary intake [21]. Examples include urinary proline betaine (a marker of citrus fruit intake) and chlorophyll-derived metabolites like pheophorbide A and pheophytin A (marker for leafy green vegetables), which provide objective measures of metabolic effects [22,23]. These insights support the development of dietary biomarkers and targeted interventions, such as folate supplementation, to reverse adverse epigenetic changes and promote health [24,25].

DNAm and metabolomics provide complementary insights into the molecular mechanisms by which diet influences health. Metabolites, as dietary intermediates or regulators have been linked to epigenetic changes that influence gene expression and disease risk [26]. Dysregulation of these pathways is associated with metabolic disorders, cancer, and immune dysfunction [27]. Integrating the two thus allows for an wholesome assessment of dietary intake and its biological effects (Figure 1), capturing both short-term metabolic responses and long-term epigenetic adaptations [28]. This approach strengthens the evidence for diet-disease links and informs personalized nutritional interventions.

DIETARY AND METABOLITE INFLUENCES ON THE EPIGENOME

EWAS Curation

We conducted a PubMed search to identify EWASs on dietary patterns, specific foods, and diet-related metabolites using the following terms: (EWAS OR “epigenome-wide association study” OR “epigenomic”) AND (diet OR dietary OR intake OR food OR nutrition OR “dietary patterns”) AND (metabolite OR metabolomics OR “DNA methylation” OR epigenetic OR epi genomics).

The search identified 700 manuscripts in PubMed, covering human studies published between 2006 and 2025 (as of February 24, 2025). To ensure methodological consistency, we excluded studies published before 2011, prior to the launch of the Illumina HumanMethylation450K BeadChip [29], leaving 427 eligible studies. We focused on 30 of these studies that met the following criteria (Table 1): 1) Investigated associations between dietary patterns or specific foods and DNAm in human subjects; or 2) Investigated associations between diet-related metabolites and DNAm in human subjects; and 3) Utilized current-generation DNAm arrays (HumanMethylation450K or HumanMethylation EPIC); and 4) Provided sufficient detail on study population, tissue source (eg, maternal whole blood, infant cord blood), and dietary exposures or diet-related metabolite.

Data Extraction and Synthesis

For each study included, we extracted the dietary outcomes, the number of significant CpG signals, characteristics of the study population (eg, mothers, offspring), sample size, the source of DNAm data (eg, whole blood, cord blood, and array used), and details on the statistical models (Table 1). Due to the uneven sample sizes across studies, significant CpG associations were identified using each study’s multiple testing correction: either a false discovery rate (FDR) or Bonferroni, based on the primary analysis at $\alpha = 0.05$. When both were reported, we retained CpGs based on the more stringent Bonferroni correction. Whenever replication was performed, we extracted only replicated signals. For each identified signal, we extracted the CpG site, chromosome, genomic position, mapped gene, and associated outcomes (Tables S1 and S2).

Among the 18 studies focused on diet, 15 reported significant findings. Sample sizes varied widely ($n=197$ to $n=15\,789$), ranging from small cohort studies (hundreds) to large population-based meta-analyses (tens of thousands). Many studies ($n=5$) examined maternal diet and its relationship with newborn cord blood DNAm, while others ($n=13$) explored diet associations in the

general population. The primary dietary features studied included essential vitamins (eg, folate and vitamins C, E, and B12), animal and plant protein, maternal dietary glycemic index and load, dietary fatty acids, overall diet quality, tea and coffee consumption, and specific dietary patterns (eg, calorie-restricted high-fat, plant-based, omnivorous, Western, and Mediterranean diet). Emerging trends, including the impact of ultra-processed foods on DNAm, were also noted.

Among 12 diet-related metabolite EWAS studies, 11 reported significant associations. Sample sizes varied ($n=81$ to $n=3358$), though comparatively smaller than diet EWASs. Five studies reported DNAm impacted lipid-related metabolites, while one study found significant associations of DNAm with phospholipids [30-34]. Three investigations focused on the effects of maternal plasma metabolites on fetal DNAm in cord blood, all reporting significant results [35,36]. The remaining six studies focused on either specific metabolites, such as folate and metabolites associated with vitamin B intake, or a comprehensive scan of all metabolites on a panel [35-40].

In the four studies that did not report significant findings, we identified several recurring themes. First, two diet EWASs with the smallest sample sizes [41,42] were unable to detect specific CpGs associated with the targeted dietary exposures; however, they did observe relevant functional enrichment among the top signals. Second, some dietary phenotypes may require further refinement—for example, ultra-processed food intake [43], which may not be adequately captured by current assessment methods. Third, the one metabolomic EWAS [44] applied stringent corrections for all pairwise comparisons between metabolites and CpGs, which reduced statistical power.

Dietary Exposures and Epigenetic Changes

Our analysis reveals several CpG sites with overlapping associations across diverse dietary exposures (Figure 2 and Table S1). Identifying multiple exposures that map to CpGs in the same gene provides strong evidence supporting an association with dietary signals. In dietary EWASs, we found nine such signals (Table S1), the majority of which are specific to whole blood DNAm. The most robust signal is at cg03084350 (*PLCD1*), independently identified in three separate studies, on the Dietary Approaches to Stop Hypertension (DASH) diet, Vitamin C and E intake, and poor diet quality [45-47]. In the DASH diet and Vitamin C/E intake studies, lower methylation or hypomethylation at CpG sites mapped to *PLCD1* was observed with increasing intake, whereas the poor diet quality study reported higher methylation or hypermethylation. The convergence of these findings highlights common nutritional features captured across dietary exposures, particularly the nutrient-dense compo-

Table 1. A List of Studies with Significant EWAS Findings

Category	Study Author – Year	Cohorts	Sample Size	Epigenetic tissue type	Exposure	CpG as Dependent vs. Independent	Statistical Model	# of Significant CpGs
Dietary Intake	Karabegović et al. (Nat. Commun. 2021)	Rotterdam Study (Netherlands), FHS (United States), ALSPAC (UK), Atherosclerosis Risk in Communities Study (United States), Cardiovascular Health Study (United States), KORA (Germany), TwinsUK (UK), European Prospective Investigation into Cancer and Nutrition (Italy), AIRWAVE (Great Britain), ESTHER Study (Germany)	15 789	Whole blood	Coffee and tea consumption	Dependent	Age, sex, smoking status (never, former, and current), white blood cells (either measured or imputed based on the Houseman algorithm) as fixed effects, and technical covariates as random effects to control for batch effects, additionally adjusted for body mass index (BMI, kg/m ²) and alcohol consumption (g/day)	11*
	Ma et al. (Circ Genom Precis Med. 2021)	Atherosclerosis Risk in Communities study, the Framingham Heart Study, the Genetics of Lipid Lowering Drugs and Diet Network, the Multi-Ethnic Study of Atherosclerosis, the Normative Aging Study, and the Lothian Birth Cohort	6662 (European subset)	Whole blood	Mediterranean-style diet score (MDS) and the Alternative Healthy Eating Index score (AHEI)	Dependent	Age, sex, total energy intake, smoking status, physical activity, and BMI.	30*
	Schellhas et al., (Epigenetics, 2023)	Six European studies (ALSPAC, BiB, Generation R Study, MoBa Norway, INMA, EDEN Mother-Child Cohort)	3725 mother-child pairs	Cord Blood	Maternal caffeine intake	Dependent	Maternal age, maternal BMI (kg/m ²), maternal education (ordinal measure), maternal smoking during pregnancy (binary), parity (binary), estimated cord blood cell proportions (Houseman method), 20 surrogate variables for technical variation, and ethnicity (if applicable). Sensitivity models also adjusted for gestational age at birth.	1*

Dietary Intake	Ek et al. (Hum. Mol. Genet. 2017)	Four European cohorts (NSPHS, PIVUS, HWFS, EGM)	3096	Whole Blood	Coffee and tea consumption	Dependent	Age, sex, smoking status, estimated white blood cell proportions (CD8T, CD4T, NK cells, B cells, monocytes, and granulocytes).	2*
Dietary Intake	Keshawar et al. (Epigenetics. 2023)	FHS Offspring, FHS Third-Generation, UK TWINS UK, KORA, YFS, Generation 2 of the Australia-based Raine Study, Rotterdam Study, ARIC Study	11 866	Whole blood	Vitamin C and E intake	Dependent	Age, sex, body mass index (BMI), daily caloric intake, estimated blood cell type proportions (CD4+, CD8+, NK cells, monocytes, eosinophils), smoking status (never, former, current), alcohol consumption, and technical covariates. A secondary model adjusted for physical activity and diet quality.	19
Dietary Intake	Mandaviya et al., (Am J Clin Nutr, 2019)	10 European studies (Rotterdam Study, LLS, CODAM, InCHIANTI, YFS, TwinsUK, RHS, GOLDN, ARIC, CHS)	5841	Whole Blood	Dietary folate and vitamin B-12	Dependent	Age, sex, BMI, differential white blood cell counts (measured or imputed), smoking status, physical activity, B-vitamin supplement use, alcohol intake, coffee intake, and technical covariates (e.g., array number, position).	6
Dietary Intake	Domínguez-Barragán et al., (Eur J Prev Cardiol, 2023)	Four European Studies (REGICOR, FOS USA, Womens Health Initiative, AIRWAVE)	5274	Whole Blood	3 diet quality scores	Dependent	Age, sex, total energy intake (kcal/day), smoking status (never, former, current), estimated blood cell counts (CD8T, CD4T, NK cells, B cells, monocytes, granulocytes), cohort-specific surrogate variables for technical and biological variation, ethnicity (in WHI only), and batch/chip effects (in AIRWAVE). A second model additionally adjusted for BMI (kg/m ²).	18*
Dietary Intake	Do et al. (Int J Epidemiol. 2021)	WHI, TwinksUK	4926	Whole blood	Diet quality	Dependent	Cell composition, chip number, chip location, study characteristics, principal components of genetic relatedness, age, smoking status, race/ethnicity, and body mass index (BMI).	24

Dietary Intake	Leorguillé et al., (J Nutr, 2023)	Lifeways Cross-Generation Cohort Study (Ireland)	244 mother-child pairs & 130 father-child pairs	Cord blood	Parental diet quality	Dependent	Child sex, parental smoking status, batch effect (sample plate), estimated cell-type proportions (buccal epithelial cells, B cells, granulocytes, natural killer cells), maternal age, education level, parity, birthweight, maternal BMI, gestational age at birth, and energy intake (for DASH models only).	2*
Dietary Intake	Laurado-Pont et al., (Clin Epigenetics, 2025)	Four European studies (HELIX, Generation XXI, ALSPAC, and Generation R)	3152 children	Whole Blood	Ultra-processed food consumption	Dependent	Child age, sex, ethnicity (or HELIX sub-cohort), maternal age, maternal early-pregnancy BMI, maternal smoking status, maternal education level, child BMI, child sedentary behavior (physical inactivity), and fruit and vegetable intake.	0
Dietary Intake	Hellbach et al., (Eur J Nutr, 2022)	KORA FF4, TwinsUK, and Leiden Longevity Study	2293	Whole blood	37 dietary exposures	Dependent	Sex, age, age ² , BMI, BMI ² , total energy intake, alcohol intake (where applicable), smoking status, estimated white-blood-cell proportions, and methylation batch/plate (plus surrogate variables in KORA).	49
Dietary Intake	Küpers et al. (Diabetes Care 2022)	ALSPAC (UK), Generation R Study (Netherlands, INMA (Spain)	2003 mother-child pairs	Cord blood	Maternal dietary glycemic index and load	Dependent	Age, sex, BMI, smoking status, white-blood-cell proportions, technical batch effects, physical activity, total energy intake, estrogen-therapy use, and PUFA supplement intake.	3*
Dietary Intake	Lange de la Luna et al., (Clin Epigenetics, 2024)	KORA FF4 study (N = 1354) and LLS (N = 448)	1802	Whole blood	up to 8 dietary fatty acid intake	Dependent	Age, sex, body mass index (BMI), smoking status (current/former/never), white blood cell proportions (monocytes, basophils, eosinophils, neutrophils, lymphocytes), sample plate number (batch), physical activity, total energy intake, estrogen-therapy use, and PUFA supplement intake.	3*

Dietary Intake	Ott et al., (Diabetes Care, 2023)	Five European studies (INMA, Leipzig Atherosclerosis Cohort, The LIFE Child Study, TEENLIFE AB, Northern Finland Birth Cohort, Raine Study Gen2 (Australia))	1187 children/adolescents	Whole blood	Dietary Glycemic Index and Glycemic Load	Dependent	Age, sex, parental education level, smoking, total energy intake, blood cell-type proportions, technical batch variables (e.g. plate or array), and any cohort-specific covariates (e.g. study centre).	1
Dietary Intake	El Sharkawy et al. (Epigenetics 2024)	CHOP Trail (Germany, Belgium, Italy, Poland, and Spain), Generation R Study (Netherlands)	1183 children	Whole Blood	Animal and plant protein intake during infancy	Dependent	Child sex, study centre (Nancy vs Poitiers), maternal age at delivery, smoking during pregnancy (never, 1–9 cigarettes/day, ≥ 10 cigarettes/day), gestational age, batch/plate effects, estimated placental cell-type proportions, surrogate variables (SVA), and vitamin supplementation before and during pregnancy.	4
Dietary Intake	Küpers et al. (Epigenetics, 2022)	ALSPAC, Generation R Study, INMA, Healthy Start, Project Viva	2802 mother-child pairs	Cord blood	Adherence to the Mediterranean diet	Dependent	Child sex, maternal educational level, maternal age, maternal smoking, maternal body mass index, maternal total energy intake, batch (technical), estimated cell-type proportions	1*
Dietary Intake	Lecorguillé et al. (Epigenetics 2021)	EDEN mother-child (France)	573 mother-child pairs	Placental tissue	Dietary patterns (varied and balanced diet, 'vegetarian tendency,' and 'bread and starchy food')	Dependent	Child sex, maternal age at delivery, study centre (Poitiers vs Nancy), smoking during pregnancy (none / 1–9 cigarettes / ≥ 10 cigarettes per day), gestational age at birth, batch (array plate), estimated placental cell-type composition (RefAC-Tor PCs), surrogate variables (SVA), vitamin supplementation before/during pregnancy	0*
Dietary Intake	Ramaker et al. (J Gerontol A, 2022)	CALERIETM Trial (United States)	197	Whole blood	Calorie restriction (CR) over 12 and 24 months	Dependent	Age, sex, body-mass-index stratum (22.0–24.9 vs. 25.0–27.9 kg/m ²), study site, genetic-ancestry principal components 1–3, estimated blood-cell proportions (monocytes, neutrophils, CD4 ⁺ T cells, CD8 ⁺ T cells, natural-killer cells, B cells), EPIC-array control-probe principal components 1–7	0

Metabo- lites	Costeira et al. (Clin Epi- genetics 2023)	Two European studies (TwinsUK, KORA)	3358	Whole blood	18 metabolites were associated with B vitamin intakes	Dependent	Age, smoking status during pregnancy, estimated blood- cell proportions, ancestry (as a random intercept)	16*
Metabo- lites	Joubert et al., (Nat Commun, 2016)	Two European studies (MoBa, Generation R Study)	1988	Cord blood	Maternal plasma folate	Dependent	Maternal age, maternal educa- tion level, smoking during preg- nancy, parity, methylation-array batch, array position.	48*
Metabo- lites	Battram et al., (bioRxiv, 2020)	Avon Longitudinal Study of Parents and Children (UK)	940 moth- er-child pairs	Cord blood	Maternal plasma metab- olites	Dependent	Age, top 10 ancestry principal components, and estimated cell-type proportions (CD8+ and CD4+ T cells, B cells, mono- cytes, natural killer cells, and granulocytes).	0
Metabo- lites	Bargas et al., (Epigenomics, 2023)	EIPREG (Oslo, Norway)	823	Whole blood	Maternal serum folate	Dependent	Age, smoking status, estimated blood cell proportions (CD8+ T cells, CD4+ T cells, NK cells, B cells, monocytes, granulocytes), ancestry (random intercept in cross-ancestry EWAS).	3
Metabo- lites	Sayols-Baix- eras et al. (Hum. Mol. Genet. 2016)	REGICOR (Spain), Fram- ingham Offspring Study (US)	3187	Whole blood	Serum lipid levels (total cholesterol, LDL-C, HDL-C, triglycerides)	Independent	Age, sex, smoking status (cur- rent/former/never), estimated white-blood-cell proportions (CD8+ T cells, CD4+ T cells, NK cells, B cells, monocytes, granulocytes), array number, array position	13*
Metabo- lites	Irvin et al., (Circ, 2014)	GOLDN Study, FHS Offps- ring Cohort	2252	Whole blood	Fasting very-low-density lipoprotein cholesterol and triglycerides	Dependent	Age, sex, study site, methyl- ation PC1, PC2, PC3, PC4 (to adjust for cell-purity), family kinship (random effect)	4*
Metabo- lites	Braun et al., (Clin Epi- genetics, 2017)	Rotterdam Study (Nether- lands)	1485	Whole blood	Blood lipid levels (total cholesterol, LDL-C, HDL-C, triglycerides)	Dependent	Sex, age, smoking status (cur- rent/former/never), estimated white blood cell proportions (CD8+ T cells, CD4+ T cells, NK cells, B cells, monocytes, granulocytes), array number, array position.	5*
Metabo- lites	Lai et al. (J. Lip- id Res. 2016)	GOLDN Study	979	Whole blood	Triglyceride response to a high-fat meal	Dependent	Sex, age, study site, the first four methylation-based principal components (to adjust for CD4+ T-cell heterogeneity), and family kinship (modelled as a random effect).	4*

Metabo- lites	Tindula et al., (Environ Epi- genet. 2019)	Center for the Health As- sessment of Mothers and Children of Salinas cohort (US)	81 moth- er-child pairs	Cord blood	92 maternal lipid plasma metabolites	Dependent	Child sex, batch (450K BeadChip batch), estimated proportions of cord-blood white blood cell types (granulocytes, CD4+ T cells, CD8+ T cells, B cells, monocytes, NK cells, nucleated red blood cells)	4*
Metabo- lites	Petersen et al., (Hum Mol Genet, 2014)	KORA cohort (Germany)	1814	Whole blood	649 blood metabolites	Independent	Age, sex, body mass index (BMI), white blood cell count (WBC), three SNPs within ± 5 Mb of each CpG site (in sec- ondary models)	20*
Metabo- lites	Gomez-Alonso et al., (Clin Epigenetics, 2021)	KORA F4 study (N = 1662) and replicated the results in the LOLIPOP, NFBC1966, and YFS cohorts (N = 3752)	1662	Whole blood	147 serum metabolites	Independent	Age, sex, smoking status, oral contraceptive use, first 22 principal components of DNA methylation data (non-genetic), cis-meQTLs, metaboQTLs, erythrocytes, granulocytes, basophils, eosinophils, lym- phocytes, monocytes, and thrombocytes.	16
Metabo- lites	Wu et al., (Clin Epigenetics, 2024)	LifeLines-DEEP cohort (Netherlands)	693	Whole blood	1183 plasma metabolites	Independent	Age, sex, smoking status, oral contraceptive use, first 22 principal components of meth- ylation data (non-genetic), cis- meQTLs, metaboQTLs, erythro- cytes, granulocytes, basophils, eosinophils, lymphocytes, monocytes, thrombocytes.	5

* Significance threshold using Bonferroni correction $p < 0.05$, otherwise FDR correction $p < 0.05$

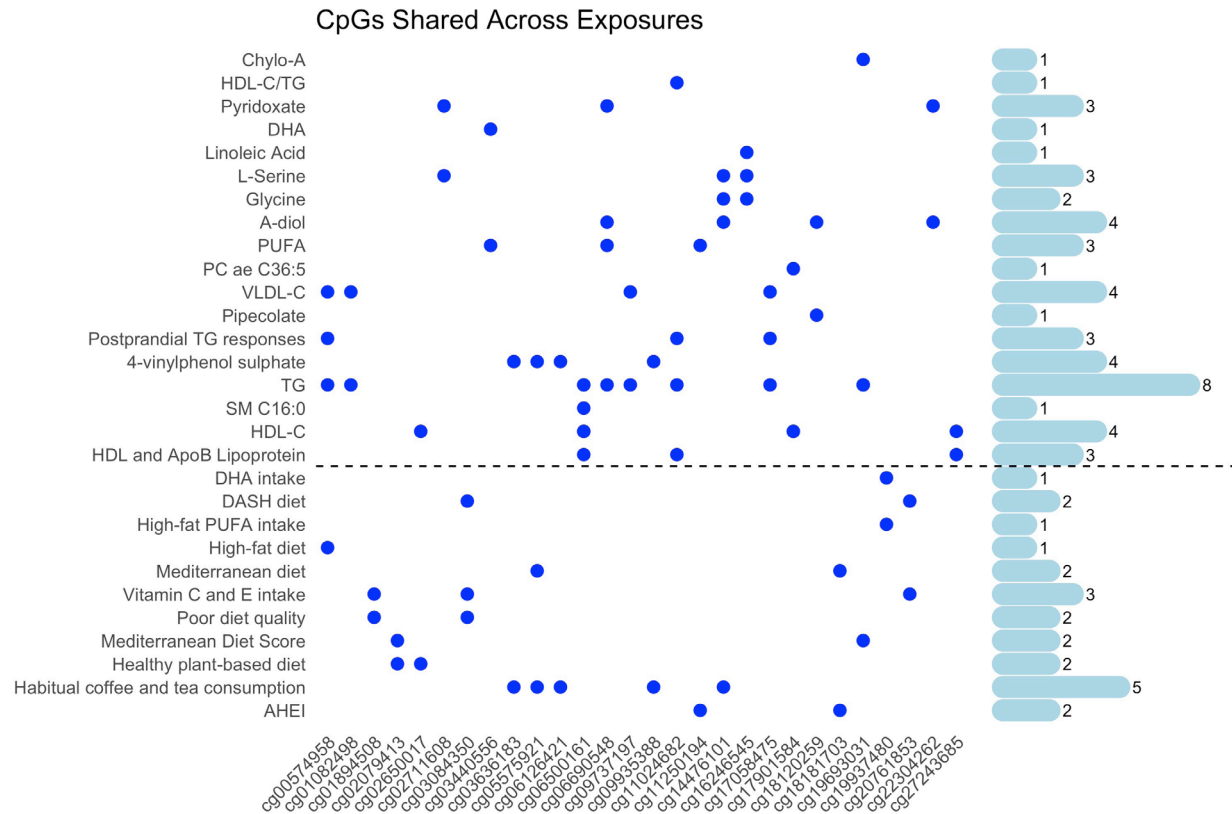


Figure 2. Common CpG Sites Associated with Multiple Dietary Exposures. This figure summarizes the significant CpG–diet associations identified in studies from Table 1. Each row corresponds to a specific dietary exposure (eg, AHEI, cream, butter), while blue dots mark individual CpG sites that show evidence of association with that exposure. Only CpGs that are associated with more than one exposure are shown to highlight shared epigenetic signals. To the right of each row, a horizontal bar represents the total number of shared CpG associations for that exposure. The bars help summarize the relative density of shared epigenetic signals across different exposures.

sition of the DASH diet, which reflects overall higher diet quality and greater intake of antioxidants such as vitamins C and E. Altogether, these findings suggest DNAm may serve as a molecular signature of antioxidant-rich, high-quality diets.

Meanwhile, cg18181703 in *SOCS3* showed associations with both the Alternative Healthy Eating Index (AHEI) and a Mediterranean diet pattern in two large independent studies with no sample overlap [48,49]. Similarly, multiple CpGs in the *AHRR* gene was observed in two independent contexts (Table S1): Mediterranean diet and habitual coffee/tea consumption [49,50]. Finally, the *COL24A1* gene (cg05781609) was identified in two independent studies, one related to butter intake [51] and another associated with poor diet quality [52].

In contrast, several signals were confined to a single study. For example, cg27344289 (*FLJ41603*) appears solely in a study on nuts-seeds intake; notably, it was replicated twice within that study [51]. Similarly, cg15928106 and cg20228731 (*FLJ43663*) are detected only within one study on habitual coffee and tea consumption, despite multiple occurrences within the same cohort [50].

Diet-related Metabolites and Epigenetic Changes

Ten robust signals were identified in the studies, which included CpGs mapped to the following genes: *PHGDH*, *TXNIP*, *DHCR24*, *SCD*, *CPT1A*, *SREBF1*, *SLC1A5*, *SLC7A11*, LOC100132354, and *ABCG1* (Table S2). We did not observe overlapping signals across whole blood and cord blood EWASs.

The most abundant epigenetic signal identified in the review was mapped to the ATP-binding cassette transporter G1 gene (*ABCG1*), which was found in six of the EWAS studies. The most robust CpG site was the cg06500161 in *ABCG1*, present in four EWAS studies using whole blood studies, and was associated with triglycerides (TG), high-density lipoprotein cholesterol (HDL-C), and C16 Sphingomyelin (SM C16:0; Table S2) [30,34,38,40]. Two of the studies reported hypermethylation at cg06500161 to increase TG levels and decrease HDL-C levels, a well-known metabolic pattern (Table S2) [30,34]. Additionally, hypermethylation at cg06500161 was reported to decrease SM C16:0, a sphingolipid that is largely present in HDL, though this was only identi-

fied in one study (Table S2) [40]. Another study found associations between TG and cg06500161 in *ABCG1*; interestingly, they found that TG may drive methylation, suggesting the possible usage of *ABCG1* as a biomarker for lipid profiling [38]. This is also supported by previous studies, which have also illustrated the role of *ABCG1* in mediating lipid homeostasis [53]. Moreover, cg27243685 in *ABCG1* was represented in two studies (Table S2) [30,38]. In one study, hypermethylation at this site was similarly associated with increased TG and decreased HDL-C levels, while the other study further linked it to increased risk of obesity.

Two additional robust CpG signals were mapped to the carnitine palmitoyltransferase 1A gene (*CPT1A*), at cg00574958 and cg17058475, identified in three separate whole blood EWASs (Table S2) [32-34]. Hypermethylation at both CpG sites was associated with decreased TG and very low-density lipoprotein cholesterol (VLDL-C). This is consistent with the role of *CPT1A* in regulating lipid transport and serves as a potential biomarker for lipid-related diseases [33,38,54]. There were several other signals in the *CPT1A* gene associated with lipid-related metabolites, but they were only identified in one study, which requires further validation (Table S2) [33].

TG was also associated with methylation at cg11024682, mapped to the sterol regulatory element-binding protein 1 gene (*SREBF1*) in four whole blood EWASs (Table S2) [30,32,34,38]. Hypermethylation at cg11024682 was shown to decrease *SREBF1* gene expression and is associated with increasing TG metabolite levels and obesity, consistent with the role of *SREBF1* as a key regulator of lipogenesis [55]. Hypomethylation at cg11024682 and increased expression of *SREBF1* was associated with higher levels of HDL-C [30], resulting in a metabolic profile similar to that observed with CpG sites at the *ABCG1* gene.

We also observed the same epigenetic signals mapping to multiple metabolites. For example, cg06690548 in the solute carrier family 7-member 11 gene (*SLC7A11*) was identified in four whole blood studies. Upon hypermethylation, studies reported decreased TG levels, obesity, sulfated steroid, and pyridoxate (Table S2) [30,37,38,40]. A similar effect has been observed at two CpG sites at the *SLC1A5* gene, a member of the same transporter family, which may strengthen the reliability of these associations [37,40]. Another example was cg17901584 at the 24-dehydrocholesterol reductase gene (*DHCR24*), as reported in three separate studies [56-58]. Hypermethylation at cg17901584 was associated with increased HDL-C, decreased myocardial infarction (MI), and increased PC ae C36:5, a glycerophospholipid found in cell membranes (Table S2) [34,38,40]. Similarly, three studies identified hypermethylation at cg19693031 (*TXNIP*) was associated with a decrease in TG, diabetes risk,

and Chylo-A (Table S2) [30,38,40].

Many epigenetic signals linked to diet-related metabolites were also driven by serum folate (vitamin B9) and vitamin B6, which have a key role in methylation processes [36]. Serum folate is correlated with dietary folate [59], which is found in green leafy vegetables or folic acid supplementation [60]. There were 51 CpG sites strongly associated with maternal folate in two EWAS studies, one based on infant cord blood and the other one maternal whole blood, but no signal overlap (Table S2) [36,61]. Another whole blood study reported 16 CpG sites linked to pipecolate, a metabolomic biomarker for vitamin B6 and folate intake, pyridoxate, a direct catabolic product of vitamin B6, and docosahexaenoate (DHA), an omega-3 fatty acid high in cold-water fish [37]. Two studies reported a signal at cg03440556 with stearoyl-CoA desaturase (*SCD*), where one study found hypomethylation to be associated with decreased DHA and another study found it associated with increased SCD (Table S2) [37,38].

Finally, three whole blood studies identified two signals at the phosphoglycerate dehydrogenase gene (*PHGDH*); however, they reported different outcomes. Hypermethylation at cg16246545 was reported to be associated with increased glycine and L-serine in one study while it was observed to be associated with decreased risk of obesity in another (Table S2) [38,39]. Hypermethylation at cg14476101 (*PHGDH*) was found to be associated with decreased 17beta-diol disulfate, a cholesterol molecule, and increased glycine and L-serine [39,40].

IDENTIFYING OVERLAPPING EPIGENETIC SIGNALS IN DIET AND METABOLITE EWAS

We summarized significant CpG-diet related associations in Figure 2, showing the 28 CpGs that were associated with more than one exposure across 11 dietary exposures and 18 metabolites. Of these 28 CpGs, only nine were shared between diet and diet-related metabolites, while the remaining 19 were shared within diet exposures or within metabolites only. Across all exposures, habitual coffee/tea consumption and triglycerides showed significant associations with the largest number of unique CpGs.

CpG Sites Shared Between Diet and Metabolites

Nine CpG sites mapping to eight genes overlapped between diet- and metabolite-associated whole blood EWASs (Table 2). For example, cg00574958 in the *CPT1A* gene was associated with both high-fat diet intake and lipid-related metabolites in whole blood samples, including VLDL-C and TG. A similar pattern was observed for cg11024682 (*SREBF1*) and cg06500161 (*ABCG1*), which were associated with both high-fat diet and lip-

Table 2. Overlapping Epigenetic Signals Across Diet and Diet-related Metabolites

Associated CpG	Mapped Gene	Association with		
		Diet	Metabolite	Diseases or markers of disease
cg14476101	<i>PHGDH</i>	Habitual coffee and tea consumption	A-diols, Glycine, L-Serine	Obesity, Type 2 Diabetes, Blood pressure
cg00574958	<i>CPT1A</i>	High-fat diet	VLDL-C, VLDL-A, TG, ApoB Lipoproteins	Lipid-related disorder, BMI, obesity, type 2 diabetes, MI
cg02650017	<i>PHOSPHO1</i>	Healthy plant-based diet	HDL-C	Type 2 Diabetes, Crohn's disease
cg05575921	<i>AHRR</i>	Mediterranean diet, habitual coffee and tea consumption	4-vinylphenol sulphate	CHD, PTSD
cg03636183	<i>F2RL3</i>	Habitual coffee and tea consumption	4-vinylphenol sulphate	Lung cancer, mortality
cg06126421		Habitual coffee and tea consumption	4-vinylphenol sulphate	Lung cancer
cg09935388	<i>GFI1</i>	Habitual coffee and tea consumption	4-vinylphenol sulphate	Lung cancer
cg11250194	<i>FADS2</i>	Alternative Healthy Eating Index	Poly-unsaturated fatty acid	TG concentration
cg19693031	<i>TXNIP</i>	Mediterranean diet	Triglyceride and Chylo-A	Type 2 Diabetes, CVD

id-related metabolites, including HDL-C and TG. HDL-C is found in foods containing omega-3 fatty acid such as salmon and flaxseeds [62]. In contrast, cg02650017 (*PHOSPHO1*) showed an overlap between plant-based diets and HDL-C levels. Epigenetic modifications at *PHOSPHO1* might be a key link between plant-based diets and improved lipid profiles. Similarly, there was an overlap between Mediterranean diet and lowered TG and Chylo-A levels at CpG site, cg19693031 (*TXNIP*), also associated with decreased risk for type 2 diabetes (T2D) and cardiovascular disease (CVD). This further reinforces the idea that Mediterranean diet promotes a healthier lipid profile, potentially mediated by epigenetic mechanisms.

Another unexpected overlap emerged in CpG sites linked to both habitual coffee and tea consumption and 4-vinylphenol sulphate, a metabolite associated with smoking exposure. cg05575921 (*AHRR*), cg03636183 (*F2RL3*, *GFI1*), cg06126421 (*HDAC4*), and cg09935388 (*GFI1*) all exhibited methylation changes in response to both dietary intake and smoking-related metabolites. These CpGs have been well-documented in smoking exposure studies, yet their presence in relation to coffee and tea consumption suggests either a potentially shared pathway or confounding [63]. While one possible explanation is that polyphenols or other bioactive compounds in coffee and tea influence detoxification pathways similarly to tobacco exposure, the more likely explanation is confounding as previous studies have found a positive

association between smoking and caffeine consumption [64].

Collectively, these overlapping CpG sites illustrate that dietary patterns and metabolite levels do not operate independently but instead influence DNAm in ways that reflect broader metabolic processes. Indeed, we found moderate enrichment of KEGG pathways [65,66], in fatty acid metabolism and PPAR signaling pathways (Figure 3A; Table S3; adjusted *p*-value < 0.1). Additional pathways include alpha-linolenic acid metabolism, biosynthesis of unsaturated fatty acids, glycine/serine/threonine metabolism, and fatty acid degradation, though each represented by single genes. This suggests that CpGs responding to both dietary exposures and metabolite levels are concentrated in core lipid metabolic processes. These converging findings also strengthen the biological plausibility of diet–epigenome interactions and highlight candidate pathways for further mechanistic and experimental investigation.

Distinct Epigenetic Signatures and Plausibility

Many CpG sites appear in only one study, making it unclear whether they reflect true biological effects or spurious signals. The full list of CpGs identified is available in Tables S1 and S2. While mapped genes from diet-specific CpGs were not enriched in any pathways, the metabolite-unique signals show a broader metabolic profile, retaining the fatty acid metabolism and PPAR

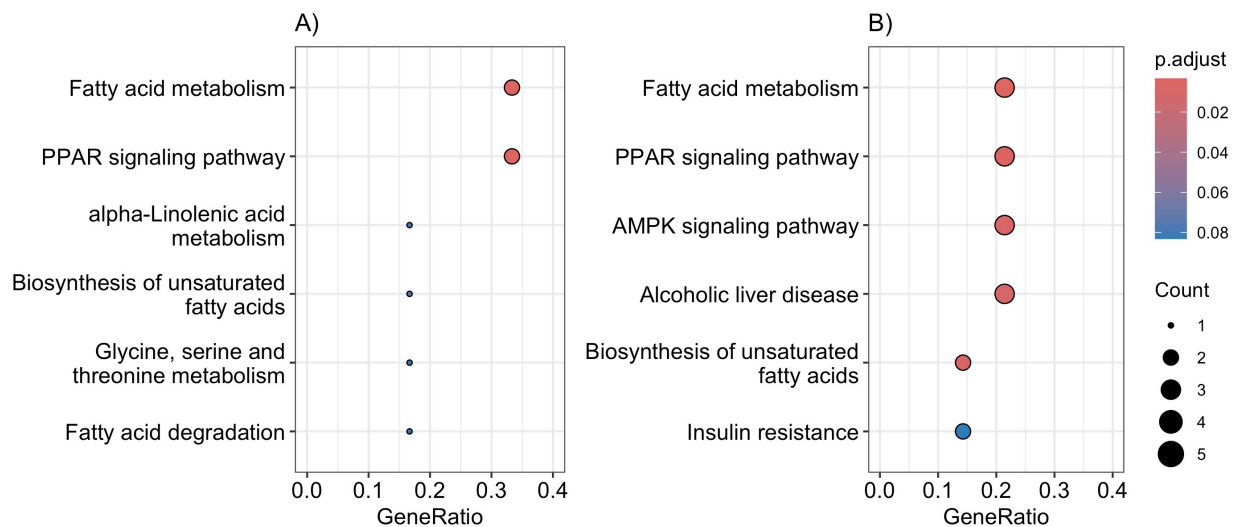


Figure 3. KEGG pathway enrichment analysis for genes identified from nutritional EWASs. Dot plots show the top enriched KEGG pathways based on gene ratio identified in (A) both dietary and metabolite EWASs, or Table 2 and (B) metabolite EWASs, or Table S2. The x-axis represents the Gene Ratio (the proportion of input genes associated with each pathway). Dot color indicates the adjusted p -value ($p.adjust$), with red denoting higher significance. Dot size reflects the number of genes mapped to each pathway. Pathways such as Fatty acid metabolism and PPAR signaling pathway were enriched in both groups, whereas unique enrichment patterns (eg, Insulin resistance) highlight additional biological processes linked to epigenetic signals of metabolites.

signaling pathways but adding energy regulation (AMPK signaling) and disease-related pathways (alcoholic liver disease, insulin resistance; Figure 3B). The AMPK pathway enrichment is particularly notable as it represents cellular energy sensing. The inclusion of pathological pathways like alcoholic liver disease and insulin resistance suggests these metabolite-unique CpGs may be more directly linked to metabolic dysfunction.

While some methylation changes align with expected biological pathways, while others suggest confounding or methodological limitations. Inconsistencies, such as disease-associated methylation changes appearing in protective dietary patterns, indicate lifestyle factors like smoking, medication use, or socioeconomic status may be influencing associations rather than diet itself. Further, recent studies highlight the role of dietary fats, particularly saturated and trans fats, suggesting that their health effects depend on dietary context rather than absolute intake [67].

Notably, some expected overlaps were absent. For instance, since nuts and vegetable oils are key components of plant-based and Mediterranean diets, one might anticipate convergence between signals from these specific exposures and those from broader plant-based dietary patterns. However, CpG sites associated with nuts-seeds do not overlap with signal for healthy plant-based diets or with the plant-oils signal. This lack of convergence suggests that the methylation profiles for individual dietary

components remain distinct from those reflecting overall dietary patterns, potentially reflecting the complexity of diet, methodological differences across studies, or confounding from related lifestyle factors. Additionally, the lack of concordance may be due in part to false-positive findings in the literature, arising from limited sample sizes or imprecise dietary recall in some studies [68].

Several associations likely reflect residual confounding rather than direct effects of diet. For example, CpG sites such as cg05575921 (*AHRR*) and cg03636183 (*F2RL3*) are robustly associated with tobacco exposure [69,70], yet they appear in studies of coffee and tea intake, behaviors often correlated with smoking [71,72]. While smoking status was adjusted for in both instances, DNAm changes due to light or past smoking can still be detected at these CpGs [73]. In this case, residual confounding may persist due to unmeasured variation in smoking intensity, duration, or recency, without quantitative metrics (eg, pack-years, cotinine levels) or methylation-derived smoking scores.

While many studies attempt rigorous adjustment (Table 1), there is considerable heterogeneity in covariate selection, statistical modeling, and tissue sources, limiting comparability and interpretability across findings. Moreover, only a handful of studies conduct sensitivity analyses to explore potential mediation or bi-directional relationships. We highlight the evolution in confounder adjustment practices across the literature, noting greater

thoroughness of covariates included in recent dietary-related EWASs. Covariate adjustment consistently included age, sex, smoking status, and tissue-specific estimated cell proportions, with many also adjusting for body mass index (BMI) and total energy intake. However, there was notable heterogeneity in the inclusion of lifestyle, dietary, and socioeconomic factors, as well as in the handling of technical variation. Recent studies have improved in incorporating ancestry principal components and batch correction methods, but gaps remain in adjusting for metabolic traits and comprehensive lifestyle exposures.

HEALTH IMPLICATIONS OF DIET-METABOLITE EPIGENOME INTERACTIONS

Evaluation of Association with Diseases

To evaluate the broader relevance of CpG sites associated with dietary exposures, we mapped the identified diet-related CpGs to the EWAS catalog [44] and EWAS Atlas [74] by systematically searching for putative associations with metabolic, cardiovascular, and other disease-related traits. EWAS Catalog and EWAS Atlas are publicly available databases of published and unpublished EWASs. To capture potential signals that may not meet the strict epigenome-wide significance threshold ($\alpha = 3.6 \times 10^{-8}$) [75], we applied a lenient yet more stringent threshold than the suggested threshold for the 450K array ($\alpha = 2.4 \times 10^{-7}$), setting our cutoff at $p < 1 \times 10^{-7}$. This reflects the exploratory nature of our study, allowing for the identification of biologically plausible overlaps between diet-related CpGs and disease traits that may otherwise be missed.

Metabolic and Cardiovascular Disease Risk

The majority of the CpGs identified in metabolite EWASs show cardiovascular associations due to the large number of signals from lipid-related metabolites, see Table S2. For example, CpGs at the *ABCG1* gene are associated with elevated triglyceride levels and reduced HDL-C, is a hallmark of insulin resistance and is commonly observed in individuals with metabolic syndrome (MetS), T2D, and diets high in refined carbohydrates. Indeed, we found CpGs at the *ABCG1* gene are associated with MetS, BMI, and waist circumference, reinforcing its regulatory role in controlling TG and HDL-C. In fact, it has been demonstrated recently that individuals with lower methylation at the *ABCG1* gene experienced greater reductions in body fat distribution after an average-protein diet intervention [76]. Hypermethylation at CpGs in the following genes: *PHGDH*, *DHCR24*, *CPT1A*, and *SLC7A11*, were linked to decreased BMI and waist circumference. These are consistent with the observation that hypermethylation at these sites decreases harmful lipids

and increases HDL-C, suggesting a protective metabolic effect. In contrast, hypermethylation at *SREBF1* is likely positively associated with BMI and waist circumference, consistent with its potential role in lipid accumulation and the risk of cardiovascular-related diseases. DNAm at *TXNIP* may be associated with a reduced risk of MetS, as multiple studies have reported it to be associated with decreased lipid levels. Interestingly, *SLC1A5* was linked to BMI, although there is limited evidence supporting a role for DNAm at this site in lipid regulation. Given the close link between lipid metabolism and adiposity, epigenetic markers that influence lipid levels are often expected to correlate with BMI, making this association noteworthy but not yet mechanistically clear.

Further, CpGs in the *PHGDH* gene that are associated with habitual coffee/tea consumption and multiple metabolites were also reported to have associations with blood pressure and hypertensive diseases. Circulating amino acids and DNAm levels at *PHGDH* were also found in previous studies to be negatively correlated with BMI [77], further supporting the role of *PHGDH* in regulating both protein-based and fat-based metabolites. It also highlights *PHGDH*'s protective role against metabolic disorders such as obesity and T2D. It has also been recently interrogated as a potential epigenetic marker to explore gene-diet interaction, and there is evidence that baseline DNAm at the *PHGDH* gene may modify how individuals respond to dietary interventions for blood pressure control [78].

Several epigenetic markers identified in our study were consistently associated with metabolic traits, irrespective of the specific dietary exposure (Table S1). For example, *PLCD1* (cg03084350) was linked to metabolic outcomes such as altered cholesterol parameters and indicators of cardiovascular risk. cg20761853 (*TIMP2*) is another example; it appears in association with Vitamin C/E intake and the DASH diet, and in both cases, the associated traits point to improved metabolic homeostasis. We found healthier diets tend to produce a hypomethylated profile at *TIMP2*, which is associated with better lipid and glucose regulation. Conversely, markers like *SLC16A3* (cg01944226 and cg08429256) consistently show hypermethylation in settings linked to poor metabolic outcomes, such as MetS and T2D. Likewise, *AHRR* (cg05575921) and *COL24A1*—although originally noted for overlapping dietary exposures—are associated with adverse metabolic traits, such as elevated triglyceride levels, impaired glucose regulation, and increased inflammatory markings [79].

Diet-induced epigenetic changes have been implicated in several other metabolic diseases. In NAFLD, high-fat and high-sugar diets may alter methylation in genes regulating lipid storage and liver inflammation [80,81]. Together, these relationships suggest that diet may shape

disease risk through epigenetic regulation across multiple metabolic pathways. Together, these relationships suggest that diet may shape disease risk through epigenetic regulation across multiple metabolic pathways.

Chronic Diseases and Others

Cord blood DNAm for cg23291200, cg19870717, cg16613938, within adenomatous polyposis coli 2 (*APC2*), were found associated with maternal plasma folate [82]. *APC2* is known to be involved in brain development and cancer etiology as a tumor suppressor gene [83]. Additionally, the same study observed various CpGs mapping to metabotropic glutamate receptor 8 (*GRM8*), which is linked with attention-deficit hyperactivity disorder and autism spectrum disorder. This further reinforces folate's role as an epigenetic marker for congenital disorders and possibly early indicators for developmental abnormalities.

Several epigenetic signals are reported to be associated with inflammatory and immune-related outcomes. *ASPRV1* (cg01894508) is particularly notable as it is detected in studies of poor diet quality and lower Vitamin C/E intake [52,84]. Hypermethylation at *ASPRV1* correlates with elevated C-reactive protein levels, systemic lupus erythematosus, and mild cognitive impairment. Similarly, while *AHRR* (cg05575921) is classically recognized as a smoking biomarker, it has also been detected in several diet-related EWASs [49,50]. Its repeated association with traits such as impaired lung function and cognitive decline (Table S1), both of which can be influenced by dietary inflammatory load, suggests that methylation at the *AHRR* gene may reflect broader environmental exposures related to inflammation. In addition, *PRDX1* (cg19937480) is observed in relation to both docosahexaenoic acid (DHA) intake and high-polyunsaturated fatty acid (PUFA) consumption [85]. Although DHA is a subclass of PUFA, the observation that *PRDX1* appears consistently across studies assessing both specific (ie, DHA) and broader (ie, PUFA) intake patterns supports this CpG as a nutrient-responsive epigenetic site linked to myopia risk. Together, these overlapping signals, where multiple, distinct dietary assessments yield similar health outcomes reinforce the biological plausibility that these epigenetic modifications are integral to the diet–disease relationship.

KNOWLEDGE GAPS AND FUTURE DIRECTIONS

Despite advances in diet-metabolome-epigenome research, key gaps remain in study designs, data sources, and translational impact. Below, we highlight some limitations and suggest directions for future research.

Need for Longitudinal Studies to Capture Temporal Dynamics

Longitudinal studies are necessary to capture the dynamic and temporal effects of diet on epigenetic modifications and metabolic processes [86]. Longer exposure periods are often needed to detect consistent and biologically meaningful patterns in observational studies, whereas metabolites and epigenetics are dynamic. For example, birth weight epigenetic signatures diminish by age 5, no longer associated with later life body weight [87,88]. With the increasing number of CpGs linked to specific dietary exposures, methylation profile scores [89] may prove useful for monitoring the cumulative epigenetic impact of dietary exposure over time, as opposed to epigenetic aging [90], which reflects global patterns and is more susceptible to lifestyle factors beyond diet. Longitudinal studies will help distinguish transient metabolic changes from epigenetic alterations that contribute to chronic disease risk, metabolic health, and aging-related processes.

Expanding Multi-Omics Approaches for Causal Insights

While diet-related metabolites have been suggested to be biomarkers of dietary intake, their reliability can be inconsistent due to multiple biological and environmental factors [91]. While some metabolites strongly reflect dietary consumption [92], others can be influenced by individual metabolism, gut microbiota composition, genetic variation, and lifestyle factors, leading to variability in their association with actual dietary intake [93]. To fully understand the role of metabolites in the diet-epigenome relationship, studies that simultaneously assess dietary intake, circulating metabolites, and DNAm are needed, allowing for robust mediation analyses that distinguish direct dietary effects from metabolite-driven epigenetic modifications. Further, as evidenced by the epigenetic signals associated with tea and coffee consumption, we observed overlap with previously reported signals for smoking. This highlights the need for caution when interpreting dietary EWAS findings, as such analyses remain susceptible to confounding factors, much like traditional epidemiological studies of diet.

A persistent limitation across nutritional EWASs is the inconsistent adjustment for key confounders, including smoking, BMI, physical activity, and socioeconomic status, and also DNAm specific confounds, such as cell compositions and batch effects. These variables can be correlated with dietary exposure and DNAm, complicating interpretation. For instance, our review found smoking-related CpGs such as cg05575921 (*AHRR*) appearing in studies of coffee and tea consumption, potentially reflecting behavioral confounding rather than dietary ef-

fects. Similarly, epigenetic changes at genes like *ABCG1* and *SREBF1*, frequently linked to lipid metabolism and obesity, may reflect adiposity-driven changes if BMI were not adequately controlled. Standardized covariate adjustment procedures are needed to clarify whether diet plays a direct role in these methylation signatures.

Finally, there is growing evidence suggesting that certain metabolites can regulate epigenetic modifications [94], but much of it remains based on associations rather than definitive causal relationships. Emerging statistical approaches such as epigenome-wide Mendelian randomization [95] and causal inference models could help establish the causal role of epigenetic modifications in disease development. Multiple data layers, including diet, metabolomics, epigenomics, and perhaps transcriptomics and proteomics, will provide the foundation to identify biologically relevant pathways and explore causal mechanisms.

Addressing Tissue-Specificity in Diet-Epigenome Research

Another key gap in current research is the lack of tissue-specific investigations in nutritional EWAS. For instance, the role of maternal circulating metabolites in shaping DNAm in newborn (eg, cord blood) remains largely unexplored, yet it may serve as a key mediating mechanism in early-life health programming, aligning with the core principles of the Developmental Origins of Health and Disease (DoHaD) framework [96-98]. Further, DNAm patterns in blood may not reflect changes occurring in metabolically active tissues [99], such as the liver, skeletal muscle, or adipose tissues. Future research should incorporate multi-tissue epigenetic profiling to determine whether diet-induced DNAm changes are systemic or tissue-specific – as was done for epigenetic clocks [88,100-105]. Finally, advances in liquid biopsy techniques [106] and single-cell resolution epigenetic analysis [107] may provide more targeted insights into diet-related epigenetic modifications in specific cell types.

Improving Ethnic Diversity in Nutritional EWAS

Another pressing need is to improve ethnic diversity in diet-metabolome-epigenome research. While some studies and meta-analyses have made efforts to include non-European populations, the majority of nutritional EWASs still come from White European cohorts. For example, dietary patterns rich in refined sugars, starches, or fiber may have distinct epigenetic effects across populations due to differences in insulin sensitivity, gut microbiota composition, and lipid metabolism. Expanding research to include ethnically diverse populations is needed to identify population-specific epigenetic and metabolic responses to diet and ensuring that dietary guidelines are

inclusive and relevant across different genetic and cultural backgrounds.

Translating Diet-Epigenome Findings into Public Health and Precision Nutrition

Research triangulating diet-metabolome-epigenome is rapidly expanding; however, its clinical and public health translation remains limited. Key challenges include determining whether epigenetic biomarkers are reversible through diet and whether they can inform risk prediction or personalized nutrition. The majority of current studies focused on identifying CpG sites are strongly associated with both diet and metabolic outcomes, but few uses causal inference or randomized control trials [16,108,109]. Thus, the effectiveness of interventions (eg, dietary changes, supplements, lifestyle modifications) in altering disease trajectories remains to be established. Dietary intervention trials [110,111], incorporating epigenetic and metabolomic assessments, will be essential for determining whether modifying dietary intake can reverse adverse epigenetic changes and improve health outcomes.

CONCLUDING REMARKS

In this review, we identified nine diet-metabolome-epigenome associations and summarized the current state of EWASs literature on diet-related exposures. The shared diet-metabolite CpGs appear focused on fundamental fatty acid processing, while metabolite-unique signals extend into broader metabolic regulation and pathological states. Understanding how dietary exposures relate to epigenetic changes through metabolic pathways may inform future nutritional guidelines, though causal pathways remain to be validated. This area of research is also in synergy with the development of personalized nutrition, emphasizing the need to move beyond one-size-fits-all recommendations. Further progress would benefit from multi-omics data, ethnic, and genetic diversity to develop more inclusive, globally relevant nutritional guidelines.

Abbreviations: EWASs, Epigenome-wide association studies; DNAm, DNA methylation; 5mC, 5-methylcytosine; CVD, cardiovascular disease; BMI, body mass index; HDL-C, High-density lipoprotein cholesterol; VLDL-C, very low-density lipoprotein cholesterol; TG, triglycerides; T2D, type 2 diabetes; MetS, metabolic syndrome; DHA, high-dose docosahexaenoic acid; PUFA, polyunsaturated fatty acid; SCD, stearoyl-CoA desaturase; CpG, cytosine-phosphate-guanine; AHEI, Alternative Healthy Eating Index.

Appendix A: Supplementary Tables

Table S1. CpGs significantly associated with dietary intake.

Table S2. CpGs significantly associated with diet-related metabolites.

Table S3. Enriched KEGG Pathways for Diet and Metabolite-Associated CpG-Mapped Genes.

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